



GATE CS / IT 2027

Theory of Computation

Quick Revision Sheet

Complete GATE syllabus coverage — All topics and subtopics

Chomsky Hierarchy

Type	Language	Machine	Closed Under
Type 3	Regular	DFA/NFA	Union, Concat, Star, Complement, Intersection
Type 2	Context-Free	PDA	Union, Concat, Star (NOT complement/intersection)
Type 1	Context-Sensitive	LBA	All boolean ops
Type 0	Recursively Enum.	TM	Union, Concat, Star

Finite Automata

DFA & NFA

- DFA: 5-tuple $(Q, \Sigma, \delta, q_0, F)$, deterministic
- NFA to DFA: subset construction, max 2^n DFA states
- e-NFA: NFA with epsilon transitions
- Epsilon-closure: states reachable via epsilon only
- Both DFA and NFA recognize exactly regular languages

Minimization & Pumping

- Table-filling: mark pairs (q, r) where $q \in F, r \notin F$
- Merge unmarked pairs $\rightarrow$ minimal DFA
- Myhill-Nerode: min states = equivalence classes
- Pumping Lemma (Regular): $w=xyz, |xy| \leq p, |y| \geq 1, xy^iz \in L$
- Use pumping lemma to prove non-regularity

Context-Free Languages

CFG & PDA

- $G = (V, \Sigma, R, S)$: Variables, Terminals, Rules, Start
- $PDA = (Q, \Sigma, \Gamma, \delta, q_0, Z_0, F)$
- $L(PDA) = CFL$ (accept by empty stack or final state)
- CNF: every rule $A \rightarrow BC$ or $A \rightarrow a$
- GNf: every rule $A \rightarrow aB_1B_2...B_k$

CFL Properties

- Pumping Lemma CFL: $w=uvxyz, |vxy| \leq p, |vy| \geq 1, uv^ixy^iz \in L$
- $a^n b^n c^n$ is NOT CFL (use pumping lemma)
- CFL is closed under: union, concatenation, Kleene star
- NOT closed under: intersection, complement

Turing Machines & Decidability

Decidability

- Decidable: TM always halts (accepts or rejects)
- RE (semi-decidable): TM accepts if $w \in L$, may loop otherwise
- Halting Problem: UNDECIDABLE - no TM decides it
- Rice's Theorem: any non-trivial property of TM language is undecidable
- Reductions: if A reduces to B, undecidability transfers

